

Uveal melanoma: management and outcome of patients with extraocular spread

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ABSTRACT

Background Extraocular spread is thought to be a negative prognostic factor on the survival in patients with uveal melanoma. Enucleation was the standard treatment in these patients. Today, depending on the size of the tumour and the type of extraocular extension, eye-preserving irradiation treatments, such as proton beams or radioactive plaques, may be employed.

Methods 2256 patients were treated between 2000 and 2007 at the Institut Curie, Paris, France for a uveal melanoma. 67 patients (3.0%) presented an extraocular extension. A retrospective study was performed to evaluate the patients' outcomes with regard to tumour recurrence and survival.

Results Eye-conserving treatment was employed in 38 (52.8%) patients. Enucleation was performed in 29 (47.2%) patients. The median follow-up was 38 (range 7 to 79) months with an overall survival rate at 5 years of 40.4% in enucleated patients and 79.3% in the eye-conserving treatment group (protons $n=19$, iodine-125 plaque $n=19$) ($p=0.01$; Kaplan–Meier analysis). No tumour recurrence was observed in any group. The degree of extraocular spread as well as the clinical characteristics tumour location, retinal detachment, ciliary body involvement ($p<0.01$; χ^2 test) and tumour thickness ($p=0.04$; χ^2 test) influenced the choice of treatment. Age, tumour diameter, involvement of optic nerve, vitreous haemorrhage and the amount of pigment did not have any influence.

Conclusions Tumour recurrence rates and survival rates were not adversely affected in patients receiving conservative eye treatment. This may thus represent a therapeutic option in certain patients with extraocular spread.

INTRODUCTION

Uveal melanoma is the most common ocular tumour, with a reported incidence of six to seven cases per 100 000 inhabitants.¹ The death rate is about 30% within 5 years.² Extraocular extension is thought to be a negative prognostic factor for orbital tumour recurrence, development of metastases and tumour-related death.³ It is reported in about 13% of enucleated eyes.^{4–7}

Enucleation was considered for a long time as the standard treatment in these patients. In recent years, eye-conservative treatment methods, including plaque and charged particle radiotherapy, were used increasingly in the treatment of uveal melanoma. They may represent an alternative treatment in selected patients with extraocular spread depending on its degree and type.^{8–14}

Our study aimed to determine tumour recurrence and survival as well to compare these findings with previously reported results on uveal melanoma with extraocular spread.

MATERIALS AND METHODS

Patients

A retrospective single-centre study was performed in patients with choroidal melanomas and cilio-choroidal melanomas with origin in the peripheral retina, presenting extraocular spread. To identify these patients a computer search was conducted on a customised database, registering patients treated for uveal melanoma at the Institut Curie, Paris, France. Patients with disseminated melanoma were excluded. Study performance and data management complied with the Declaration of Helsinki and were approved by the locally appointed ethics committee.

Recorded patient data were age, sex, best-corrected visual acuity and cancer antecedents. Clinical tumor data on location, diameter, thickness, presence of retinal detachment, amount of pigment, vitreous haemorrhage and involvement of optic nerve were equally registered. Data on extraocular extension were obtained clinically by slit-lamp biomicroscopy and B-scan ultrasonography at the initial visit (figure 1), during surgery and confirmed histologically in enucleated patients. Irradiation parameters were obtained from the radiation treatment plan.

Follow-up was performed 1 month after completed treatment, every 6 months for 2 years, and annually thereafter. A liver ultrasound for the detection of metastases was performed every 6 months for 10 years and annually thereafter.

Metastasis, local recurrence, secondary enucleation as well as information on survival were also recorded in the database.

Treatment

Plaque radiotherapy was mainly performed in tumours located anterior to the equator with the tumour height not exceeding 6 mm using the iodine-125 isotope. The plaque size selected was 2–4 mm larger than the entire tumour base. The radiation target dosage to the tumour apex was 90 Gy.

In order to position the plaque, a conjunctival peritomy was performed, sparing the area of the extraocular extension. After transillumination and episcleral marking of the tumour borders, the plaque was placed on the undisturbed conjunctiva and tenon fascia overlying the intraocular melanoma and its extraocular component.

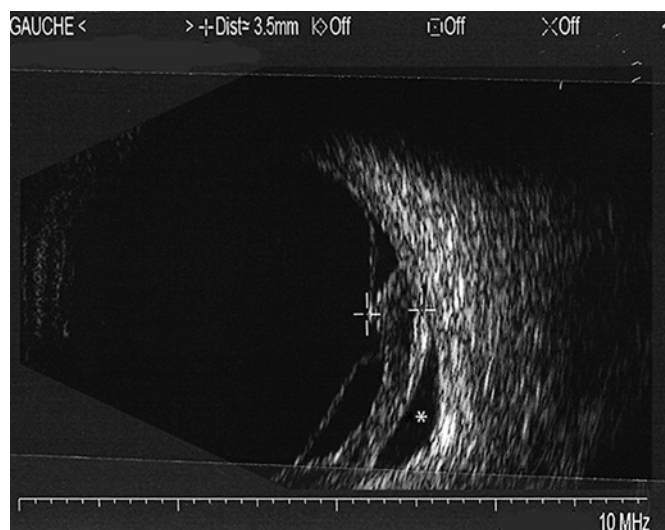


Figure 1 Ultrasound of a choroidal melanoma with surrounding retinal detachment and the extraocular extension.

Proton beam radiotherapy was performed 2 weeks after the surgical procedure for precise demarcation of tumour margins. After conjunctival peritomy, the intraocular tumour was localised by transillumination. Four to five tantalum clips (diameter 2.5 mm, thickness 0.5 mm) were positioned at the tumour margins, leaving the site of the extraocular extension untouched. Precise measurements of the size and position of the extrascleral nodule were performed.

As described previously,^{15–17} three-dimensional tumour modelling was based on clip coordinates and imaging data (fundus photography and ultrasonography). The extrascleral extension was included in the treatment plan. The EYEPLAN software was used to calculate the dose distribution adding a safety margin of 2.5 mm around the intraocular tumour and the extraocular extension.

A total dose of 60-cobalt-Gy equivalent was delivered in four fractions, with the 90% isodose placed 2.5 mm behind the tumour. Two orthogonal Polaroid x-rays verified the clip positions before each irradiation session. A 10-fold magnifying infrared camera monitored the fixed gaze allowing treatment interruption whenever the eye shifted from the original position.

During treatment, patients were immobilised in a sitting position, using a custom-made thermoplastic head mask and a mouthpiece, fixating monocularly an infrared light-emitting diode. Eyelids were excluded from the irradiation fields using retractors under topical anaesthesia.

If an extrascleral extension was detected during surgery for preparing proton beam or 125-iodine plaque radiotherapy, the

operation was, depending on the extent of extraocular spread, either continued or interrupted with closing the operation sites. For large nodules of extrascleral extension, the patient was subsequently informed of the need for enucleation, and a second surgical intervention was planned.

Enucleation was performed less than 2 weeks from diagnosis if an eye-conserving treatment was unlikely to succeed. It was completed with a snare, closing the rectus muscles over a vicryl mesh-wrapped hydroxyapatite orbital implant. After careful inspection of the globe, a tumour sample was taken by needle biopsy for subsequent cytogenetic studies. The globe was then fixed in formalin, cut and embedded in paraffin for pathohistological examination.

Postoperative orbital radiotherapy was performed with an average dose of 50 Gy over 40 days.

Statistical analysis

A χ^2 test or Fisher test in small sample sizes was used to compare two percentages. A Student t test was applied for the comparison of two means. A methodology adapted to survival analysis was used to study mortality and tumour recurrence. Survival rates were calculated according to the Kaplan–Meier method from the first day of treatment to the date of event considered (death, recurrence and metastases). A Cox-regression analysis was performed to evaluate survival according to baseline tumour characteristics. A p value of <0.05 was considered statistically significant.

RESULTS

From January 2000 to December 2007, 2256 patients with uveal melanoma were treated at the Institut Curie. Sixty-seven patients presented extrascleral tumour extension (3.0%).

The extraocular spread was diagnosed in macroscopically in 56 patients (83.6%). It was detected during the clinical examination in 36 patients (slit-lamp biomicroscopy n=28; ultrasonography or magnetic resonance tomography n=8) and in 20 patients during surgery. In 38 patients, eye-conserving radiotherapy (125-iodine plaque n=19, proton beams n=19) was performed. Eighteen patients were enucleated (enucleation n=17, exenteration n=1).

The extraocular spread was diagnosed in microscopic ways (pathology) alone in 11 enucleated eyes (figure 2).

Baseline characteristics

The baseline characteristics are listed in table 1. Thirty-four male and 33 female patients were evaluated (n=67). Their mean age was 61.6 (median 63, range 27 to 86) years.

Twenty-four melanomas were diagnosed anterior to the equator. Thirty-one tumours crossed the equator, and 12 tumours were located posterior to the equator. In more than half

Figure 2 Pathology specimen of an enucleated eye: left macroscopic extraocular extension; right microscopic extraocular extension (arrow).

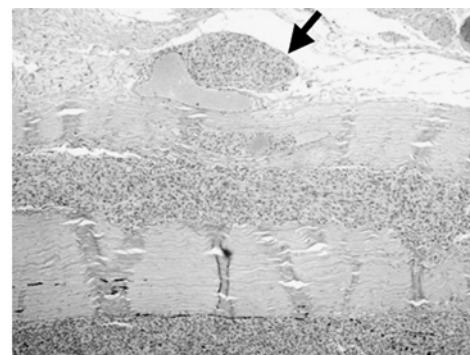
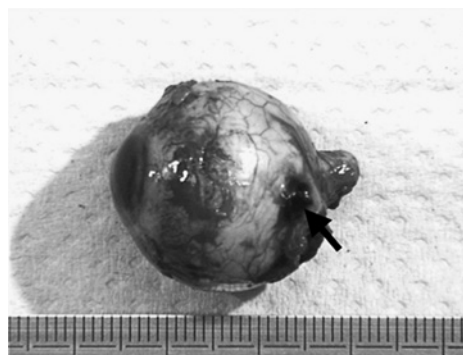
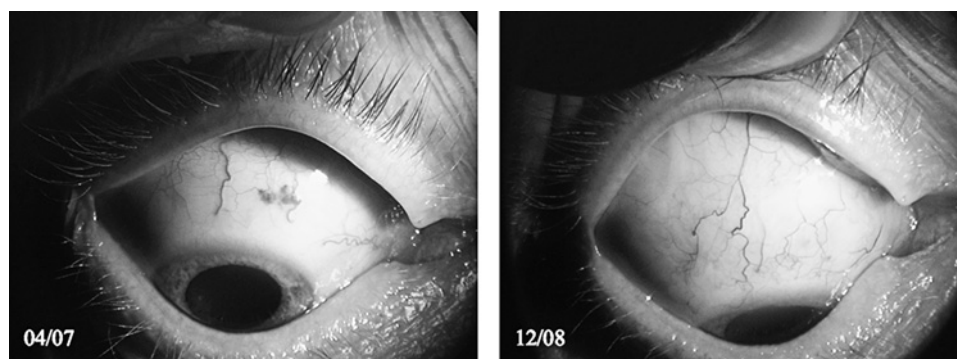


Figure 3 Anterior segment photograph presenting the extraocular extension in a patient with ciliochoroidal melanoma prior to 125-iodine plaque radiation therapy (left) and at 20 months' follow-up (right).



($n=36$), the basal tumour diameter was greater than 15 mm with a mean of 15.3 (median 16.0, range 3.0 to 25.0) mm. In 35 out of 67 patients, the tumour thickness was more than 7 mm with a mean of 7.4 (median 7.0, range 1.0 to 18.0) mm. An invasion of the optic nerve was described clinically in four patients. A retinal detachment was found in a quarter of patients ($n=17$) patients. A vitreous haemorrhage was present in two patients. An achromic lesion was observed in 12 patients.

Enucleation was the treatment of choice in 28 (41.8%) patients. In one patient, an exenteration was performed. In 38 patients (52.8%) an eye-preserving treatment was applied (125-iodine radioactive plaque $n=19$ (figure 3), proton beams $n=19$) (table 2). In one patient without light perception, a secondary enucleation was performed due to a painful globe.

Analysis of clinical baseline characteristics revealed a significant influence of tumour height, tumour location, invasion of the ciliary body and presence of retinal detachment on the decision of treatment. Twenty out of 24 patients (83.3%) with an anterior located tumour had a conservative treatment. Ten out of 12 patients with a posterior located melanoma were enucleated (83.3%) ($p<0.01$; χ^2 test). In patients with retinal detachment ($n=17$), 14 (82.4%) were enucleated ($p<0.01$; χ^2 test).

The majority of patients presenting a tumour involving the ciliary body ($n=36$) were treated by eye-preserving radiation therapy ($n=27$; 75.0%; $p<0.01$; χ^2 test). Tumour thickness also influenced the choice of treatment ($p=0.04$; χ^2 test). Patients presenting a tumour height above 7 mm ($n=32$) were more likely to be enucleated ($n=18$; 56.3%).

The degree of the extraocular extension was also important for the choice of treatment. Extraocular spread in enucleated patients (136.7 ± 346.4 , range 0.1 to 1500 mm³) was larger than in patients with conservative treatment (9.7 ± 15.2 , range 0.5 to 55.4) (table 3). All patients with an extraocular tumour larger than 1000 mm³ were enucleated (table 1).

Age, tumour diameter, clinical involvement of the optic nerve, presence of a vitreous haemorrhage or the achromic character of the lesion did not reveal a statistically significant influence on treatment decision.

We did not observe any change in the indication of treatment between 2000 and 2007 (table 2).

Pathohistological and cytogenetic data

Pathohistological results were obtained in all 29 enucleated patients (table 4).

The mean tumour diameter was 14.3 mm (median 15.0, range 3.0 to 20.0). It was not superior to the mean measured clinically for the complete patient cohort (table 1). Extraocular spread was diagnosed by microscopy alone in 11 patients (32.6%). Involvement of the ciliary body was described in seven patients (24.1%) and of the optic nerve in two patients (6.9%).

Table 1 Clinical and tumour characteristics of patients with malignant melanoma and extraocular spread

	Enucleation n = 29	Conservative treatment n = 38	Total	p Value
Age at diagnosis (years)				
Mean (range)			61.6 (27 to 86)	NS
<63 years	13	19	32	
≥63 years	16	19	35	
Gender				
Male	21	13	34	<0.01
Female	12	25	33	
Laterality				
Right eye			40	
Left eye			27	
Tumour site				
Anterior to the equator	4	20	24	<0.01
Crossing the equator	15	16	31	
Posterior to the equator	10	2	12	
Apical tumour height (mm)				
Mean (range)			7.4 (1 to 18)	
≤7	11	24	35	0.04
>7	18	14	32	
Largest tumour basal diameter (mm)				
Mean (range)			15.3 (3 to 25)	
≤15	11	21	32	NS
>15	23	17	40	
Involving ciliary body (three values missing)				
Yes	9	27	36	<0.01
No	17	11	28	
Involving optic nerve				
Yes	3	1	4	NS
No	26	37	63	
Retinal detachment (three values missing)				
Yes	14	3	17	<0.01
No	12	35	47	
Vitreous haemorrhage (three values missing)				
Yes	2	0	2	NS
No	24	38	64	
Achromic lesion (three values missing)				
Yes	7	5	12	NS
No	20	32	56	
Extraocular spread				
Mean (range)			65.6 (0.1 to 1500)	
≤1000 mm ³	27	38	65	NS
>1000 mm ³	2	0	2	

NS, non-significant.

Table 2 Treatments applied between the years 2000 and 2007

Year	Enucleation n=29	Proton beams n=19	125-iodine plaque n=19
2000 n=2	2	0	0
2001 n=9	2	3	4
2002 n=11	8	2	1
2003 n=4	1	1	2
2004 n=7	3	2	2
2005 n=11	2	5	4
2006 n=12	6	2	4
2007 n=11	5	4	2

Eight times an epithelioid cell melanoma (27.6%), nine times a spindle cell melanoma (31.0%) and 12 times a mixed-cell-type melanoma (41.4%) was found. Mitotic figures were determined in 25 cases (86.2%). A strongly increased mitotic rate with more than 22 mitotic figures was observed six times (24.0%).

Cytogenetic studies were performed in 21 out of 29 melanomas (72.4%). A monosomie 3 was described 12 times (57.1%).

Clinical course

During the median follow-up of 38 (7 to 79) months, no intraocular or orbital tumour recurrence was observed.

The overall survival rate was 95.5% (± 2.5 SD) after 1 year, 85.9% (± 4.7 SD) after 2 years and 59.9% (± 8.7 SD) after 5 years (figure 4).

During this period, 16 patients died due to the melanoma. One patient developed a second cancer, and one patient died of unknown disease (table 5).

Kaplan–Meier analysis revealed an overall survival rate at 5 years of 40.4% (± 12.6 SD) in enucleated patients compared with 79.3% (± 8.9 SD) in patients with conservative treatment ($p < 0.01$). Patients treated by proton beams showed an overall survival of 57.6% (± 16.3 SD) compared with 100% for patients with 125-iodine plaque therapy ($p = 0.01$). Cox-regression analysis including the baseline characteristics largest basal tumour diameter, tumour height, ciliary body involvement and presence of retinal detachment confirmed the significantly lower survival of enucleated patients ($p = 0.029$).

The cumulative metastasis-free survival rate (figure 5) was 88.5% (± 3.8 SD) after 1 year, 77.9% (± 5.3 SD) after 2 years, and 64.3% (± 6.7 SD) and 51.9% (± 7.9 SD) after 5 years. It was 59.0% (± 11.9 SD) in patients who underwent eye-preserving treatment and 39.4% (± 11.5 SD) in enucleated patients ($p = 0.02$).

Twenty-five patients developed distant metastases (table 6). They were located 19 times in the liver and twice in the skin. In two patients, systemic disease was identified first in the lungs. Precise information on metastatic spread was missing in a further two patients.

DISCUSSION

The retrospective analysis revealed an extraocular spread in 3.0% of melanoma patients treated at the Institut Curie. This figure is in line with previous observations, reporting a frequency between 6.2% and 14.6% in enucleated eyes^{5–7 18–20} and between 2.8 and 5.4% in patients treated by proton-beam radiotherapy.¹¹

Table 3 Size of extraocular spread in the different treatment groups applied (mm³)

	Mean $\pm$ SD (range)
Enucleation (n=29)	136.7 $\pm$ 346.4 (0.1 to 1500.0)
Conservative treatment	9.7 $\pm$ 15.2 (0.5 to 55.4)
Radioactive plaque	4.6 $\pm$ 4.8 (0.5 to 19.7)
Proton beam therapy	14.8 $\pm$ 19.9 (1.0 to 55.4)

We did not observe a lower survival rate in patients treated with proton beam or 125-iodine plaque radiotherapy. The overall 5-year survival rate in our study was 59.8% and in patients with conservative treatment alone 79.3%. Our findings thus corroborate the observations by Augsburger *et al* and Gündüz *et al*.^{8 9} They analysed patients with extraocular spread undergoing 125-iodine plaque therapy in eight and 17 patients respectively. The rate of metastases as well as the tumour-related death was reported between 62.5% and 82%.

A possible reason for the superior survival of patients with eye-conserving treatment is the possible inclusion of patients with pseudo-extrascleral tumour extension, as we did not perform any routine excisional biopsy with subsequent histopathology. A haematoma within the extraocular muscle, a benign spindle cell nevus or normal connective tissue with associated vortex vein darkening, may be suggestive of intra-vascular invasion. Scleral darkening may also presume an extraocular spread but may represent in reality a near full-thickness scleral invasion.^{21 22}

In enucleated patients, we observed a 5-year mortality of 40.4%. This figure is in the range of previously published data and confirms the preceding findings in the literature.^{4 6 19}

Augsburger *et al* and Gündüz *et al* suggested that the survival for enucleated patients and for patients with eye-conserving treatment is similar if the tumour characteristics are comparable.^{8 9} In our cohort, the tumour diameter was similar in the different treatment groups, but patients with a tumour thickness of more than 7 mm were more likely to be enucleated.

Table 4 Pathohistological and cytogenetic data of enucleated patients with malignant melanoma and extraocular spread (n=29)

Largest tumour diameter (mm)	
Mean (range)	14.3 (3.0 to 20.0)
Predominant tumour cell type	
Epithelioid	n=8
Mixed cell type	n=12
Spindle cell type	n=9
Mitotic count (per 10 high-power fields)	
Absent	n=3
1 to 10 (weak)	n=16
11 to 22 (moderate)	n=0
>22 (strong)	n=6
Missing values	n=4
Involving ciliary body	
Yes	n=7
No	n=21
Missing values	n=1
Involving optic nerve	
Yes	n=2
No	n=26
Missing values	n=1
Cytogenetics (monosomie 3)	
Yes	n=9
No	n=12
Missing values	n=8

Figure 4 Global survival of all patients (left) and for the different treatments applied (right). For all patients together, we calculated a survival rate at 5 years of 59.9% (± 8.7 SD), for enucleated patients 40.4% (± 12.6 SD) and for patients with conservative treatment 79.3% (± 8.9 SD) ($p < 0.01$).

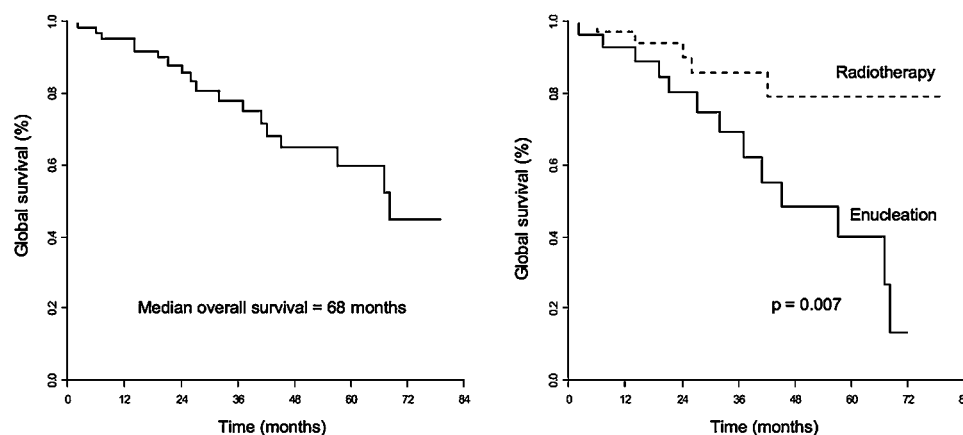


Table 5 Follow-up of patients and cause of death

Death (n=18)	
Melanoma-related death (metastases)	n=16
Second cancer	n=1
Unknown	n=1
Patients alive (n=49)	
With disease progression (metastases)	n=9
Without disease progression	n=35
Loss of follow-up	n=5

Treatment decision depended also on tumour location as well as on the extent of extraocular spread. A high percentage of patients with tumours located posterior to or crossing the equator was enucleated. Patients presenting a more voluminous extraocular extension were also more likely to be enucleated. Cox-regression analysis revealed a significantly higher mortality. However, the number of patients observed is limited. It is thus difficult to determine whether the favourable outcome of patients treated by plaque and proton beam radiotherapy is due to a more favourable mix of cases, a positive effect of treatment or a combination of these factors.

Kidd *et al* did not find any significant independent adverse effect of extrascleral extension on the prognosis for survival.²³ This may explain the similar survival of 79% in patients without extraocular spread treated by proton beams at the Institut Curie¹⁵ and the analysed patients within this study treated by conservative radiation therapy. Several authors suggested that extraocular spread is rather associated with histological features of increased local malignancy than a cause of systemic meta-

stases.^{4–7 19 24} The size of extraocular tumour and the type of scleral route seem unimportant with regard to the risk of dissemination and tumour-related death. Coupland *et al* hypothesised also that the route of extraocular spread seems to reflect only the location and the extent of the intraocular tumour.¹⁹

Pathohistological and cytogenetic results in enucleated patients revealed in more than a quarter an epithelioid cell melanoma and a monosomy 3 as markers of increased malignancy. Unfortunately, no cytogenetic data exist for patients treated with plaque or proton beam radiotherapy, so it is difficult to draw any conclusions from these analyses for the complete patient cohort. Future investigations of biopsy data will hopefully help to elucidate the significance of tumour characteristics in the analysis of patients prognosis and in the development of complementary treatment.

Being in line with our results, none of the previous studies reported any intraocular or orbital recurrence for plaque therapy.^{8 9} In enucleated patients, recurrences are nonetheless observed in about 6% with postoperative irradiation of the orbit²⁵ and 22% without.^{5 7} Within our study none of the enucleated patients developed a tumour recurrence. However, their incidence cannot be fully ascertained, as recurrences are described even 20 years or more after enucleation.^{26–28}

In conclusion, no tumour recurrence and no lower survival were observed in patients treated by iodine plaque brachytherapy or proton beam therapy. For melanomas with small degrees of extraocular extension, eye-conserving radiotherapy may thus represent an alternative therapeutic approach to enucleation. Our results support the recent hypothesis that extraocular extension may be an indicator for greater tumour

Figure 5 Metastasis-free survival for all patients (left) and for the different treatments applied (right). For the complete patient cohort, we calculated a survival rate at 5 years of 51.9% (± 7.9 SD), for enucleated patients 39.4% (± 11.1 SD) and for patients with conservative treatment 59.0% (± 11.5 SD) ($p = 0.02$).

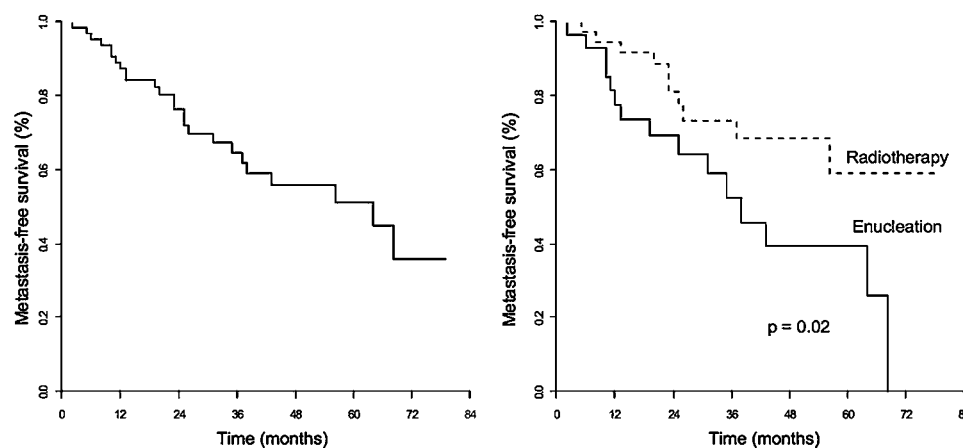


Table 6 Metastatic spread**Patients with metastatic spread (n=25)**

Liver	n=19
Skin	n=2
Lungs	n=2
Others	n=2
Patients without disease dissemination (n=42)	

malignancy.¹⁹ The patient's survival may already be destined by the time ocular treatment is started, and this awareness may have an influence on the treatment habits of uveal melanoma in the coming years.

Competing interests None.

Ethics approval Ethics approval was provided by the locally appointed ethics committee from the Institut Curie, Paris, France.

Provenance and peer review Not commissioned; externally peer reviewed.

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